Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms an equation using conservation of momentum Condone sign errors with correct terms	AO1.1a	M1	CoM $4mu - mu = mv_A + 4mv_B$ $3u = v_A + 4v_B$ NLR $v_A - v_B = 2ue$ Subtracting equations gives $5v_B = 3u - 2ue$ $v_B = \frac{u(3-2e)}{5}$
	Obtains a correct momentum equation − can be unsimplified	AO1.1b	A1	
	Forms an equation using Newton's law of restitution	AO1.1b	B1	
	Completes a rigorous argument using both conservation of momentum and the coefficient of restitution to verify the correct speed of B	AO2.1	R1	
(b)	Substitutes the speed/velocity of B back into either of their equations	AO1.1a	M1	$v_B = \frac{u(3-2e)}{5}$
	Obtains the correct speed for A − must be fully simplified	AO1.1b	A1	
(c)	Uses the maximum and minimum values of e to consider the effect on the direction of motion	AO2.4	E1	Since $0 \le e \le 1$ then expressions for the speeds above are both positive
	Deduces that the spheres both travel in the same direction after the collision and justifies their conclusion	AO2.2a	R1	Hence the spheres both travel in the same direction
(d)	Recalls the formula for impulse and substitutes a pair of corresponding velocities into the impulse formula	AO1.1a	M1	$I = 4mv_B - 4mu_B$ $I = \frac{4mu(3-2e)}{5} - 4mu$ $I = \frac{8mu(1+e)}{5}$ $0 \le e \le 1$ $\frac{8mu}{5} \le I \le \frac{16mu}{5}$
	Substitutes 0 or 1 for e into 'their' expression	AO1.1a	M1	
	Completes a rigorous argument using algebraic expressions for the velocities, the formula for impulse and the range of values for e to verify the stated inequality.	AO2.1	R1	
				Total 11 marks

Q2.

(a) $m(4u) + 3m(2u) = mv_A + 3mv_B$

M1 for four correct momentum terms with any signs.
A1 for all correct

M1 A1

$\frac{v_B - v_A}{4u - 2u} = e$

$$\begin{pmatrix} v_A + 3v_B = 10u \\ v_B - v_A = 2ue \\ 4v_B = 2ue + 10u \end{pmatrix}$$

M1 for correct terms for any signs, A1 for all correct.

M1 A1

$v_B = \frac{u}{2}(e + 5)$

OE, simplified

A1

$\left(v_A = \frac{u}{2}(e+5) - 2ue\right)$

$v_A = \frac{u}{2}(-3e + 5)$

OE, simplified

A1

6

(b) $e \leq 1 \Rightarrow v_B \leq \frac{u}{2}(1 + 5)$

Use of $e \leq 1$ (OE) needed

FT their v_B

M1

$\Rightarrow \; v_B \leq 3u$

A1

2

(c) $(I =)\; 3m.\frac{u}{2}(\frac{2}{3} + 5) - 3m.2u$

M1 for a difference of two momentums
FT their velocity from part (a)

M1

A1F for their 'Final B – Initial B'

A1F

$= \frac{5mu}{2}$ or $2.5mu$

A1

3

[11]

Q3.

(a) $v^2 = u^2 + 2as$

$v^2 = 0^2 + 2(9.8)(2.5)$

M1

$v = 7$

A1

2

(b) (i) $\frac{w}{7} = e$

M1

$w = 7e$

$0 = 7et - \frac{9.8}{2}t^2$ **or** $(0 = 7e - 9.8t)$

M1

$t = \frac{10e}{7}$ $\quad (t = 2 \times \frac{7e}{9.8})$

Answer given

A1

3

(ii) $w' = 7e^2$

$0 = 7e^2t' - \frac{9.8}{2}t'^2$

$t' = \frac{10e^2}{7}$

OE

B1

1

(c) $0^2 = (7e)^2 + 2(-9.8)h_2$

Or for correct method to find h_4

M1

$h_2 = 2.5e^2$

$h_3 = 2.5e^2$

A1

$0^2 = (7e^2)^2 + 2(-9.8)h_4$

$h_4 = 2.5e^4$

A1

$h_5 = 2.5e^4$

Total distance $= 2.5 + 2(2.5e^2) + 2(2.5e^4)$

m1

$= 2.5 + 5e^2 + 5e^4$

A1

Alternative (not in the specification)

K.E. after each bounce = $e^2 \times$ K.E. before the bounce

P.E. at max. height after each bounce = $e^2 \times$ P.E. at max. height before the bounce

(M1)

Height after first bounce = $2.5e^2$

(A1)

Height after second bounce = $2.5e^4$

(A1)

Total = $2.5 + 2(2.5e^4 + 2(2.5e^2)$

(m1)

$= 2.5 + 5e^2 + 5e^4$

(A1)

5

(d) Motion in vertical line,

No air resistance,

No energy loss,

Instantaneous bounce

B1

1

[12]

Q4.

(a) C.L.M.

$(1)3u = (1)v_A + (3)v_B$

M1 for three non-zero terms

M1A1

Restitution:

$$\frac{1}{3} \times 3u = v_B - v_A$$

Accept $v_A - v_B$

M1A1

$v_B = u$

$v_A = 0$

Solution

m1

A1 for both answers

A1

6

(b) C.L.M.

$3u = 3w_B + xw_c$

M1A1

Restitution:

$\frac{1}{3}u = w_c - w_B$

M1A1

$w_C = \frac{4u}{3 + x}$

Solution attempt, dep. on both M1s
AG

m1

$w_B = \frac{u(9-x)}{3(3+x)}$ OE

A1 for both

A1

6

(c) For further collision $\frac{u(9-x)}{3(3+x)} < 0$

M1

$9u - xu < 0$

$x > 9$

AG

A1

2

(d) $I = 5\left(\frac{4u}{3+5}\right)$

M1

$I = \frac{5u}{2}$

A1

Alternative:

$I = 3u - 3 \times \dfrac{u(9-5)}{3(3+5)}$

(M1)

$I = \dfrac{5u}{2}$

Accept $-\dfrac{5u}{2}$

Follow through on their w_B

(A1F)

2

[16]

Q5.

(a) The spheres are smooth, no force acting in **j** direction

Any valid reason

E1

1

(b) $v_A = a\mathbf{i} + b\mathbf{j}$

$v_B = c\mathbf{i} + d\mathbf{j}$

C.L.M. along **i**: $1(2) + 2(-1) = 1(a) + 2(c)$

$a + 2c = 0$

M1A1

Restitution along **i**: $c - a = 0.5(2 - (-1))$

$c - a = 1.5$

$c = 0.5$

$a = -1$

M1A1

$v_A = -\mathbf{i} + 3\mathbf{j}$

A1F

$v_B = 0.5\mathbf{i} - 2\mathbf{j}$

A1F

6

[7]

Q6.

(a) $5mu + 7mu = mv_A + 7mv_B$

Allow consistent use of positive or

negative sign for v_A.

M1A1

$12u = v_A + 7v_B$

$$e = \frac{-v_A + v_B}{4u}$$

M1

$$-v_A + v_B = 4eu$$
$$8v_B = 12u + 4eu$$

m1

$$v_B = \frac{u}{2}(e+3)$$

AG

A1

5

(b) $v_A = \frac{u}{2}(e+3) - 4eu$

M1

$v_A = \frac{u}{2}(3-7e)$

A1F

$$\frac{u}{2}(3-7e) < 0$$

M1

$3 - 7e < 0$

$$e > \frac{3}{7}$$

AG

A1

4

(c) $w_B = \frac{u}{4}(e+3)$

M1

$$\frac{u}{2}(7e-3) < \frac{u}{4}(e+3)$$

M1

$2(7e - 3) < e + 3$
$13e < 9$

m1

$e < \frac{9}{13}$

AG

A1

4

[13]

Q7.

(a) Conservation of momentum:
$0.3(3) - 0.2(2) = 0.3v_A + 0.2v_B$

M1A1

$3v_A + 2v_B = 5$ ----------------(1)
Newton's experimental law :

$0.8 = \frac{v_B - v_A}{5}$

M1

$v_B - v_A = 4$ -----------------(2)

For both (1) and (2)

A1

Solving (1) and (2)

Dependent on both M1s

m1

$v_B = 3.4$
$v_A = -0.6$

For both solutions

A1F

6

(b) $0.7 = \frac{v}{3.4}$

M1

$v = 2.38$

A1F

Speed of B (2.38) > Speed of A (0.6)
$\therefore$ B collides again with A

Cannot be gained without A1F

E1

3

[9]

Q8.

(a) conservation of momentum

$mu = mv_A + mv_B$

M1

$u = v_A + v_B$

A1

restitution

$eu = v_B - v_A$

OE

M1A1

$v_B = \frac{1}{2}u(1+e)$

OE

A1F

5

(b) $mv_B = mw_B + 2m\frac{3u}{8}$

M1A1

$ev_B = \frac{3u}{8} - w_B$

OE

M1A1

Elimination of w_B

dependent on both M1s

m1

$4e^2 + 8e - 5 = 0$

simplified quadratic equation in e *only*

A1F

$e = \frac{1}{2}$

stated as the only value (0 < e < 1 for follow through)

A1F

7

[12]

Q9.

(a) By P.C.L.M.:

$4mu = 4mv + 2\,mv$

M1

$2u = 2v + v$ ………….(1)

Law of restitution: $e = \frac{v_2 - v_1}{u}$ ……(2)

A1 for both correct

M1A1

Solving (1) and (2) →

Dependent on both Ms

m1

$v = \frac{2u - eu}{3}$

$v_2 = \frac{2u + 2eu}{3}$

A1F for both v_1 and v_2

A1F

5

(b) $\frac{12mu}{5} = 4mu - 4m\left(\frac{2u - eu}{3}\right)$

Impulse / Momentum

M1A1F

$\left[\text{or} = 2m\left(\frac{2u + 2eu}{3}\right)\right]$

$20meu = 16\,mu$

$$e = \frac{4}{5}$$

Solution to get the right answer
Answer given

A1
3

(c) $2mv = 2mv + mv$

M1

$$\frac{v_4 - v_3}{v_2} = \frac{4}{5}$$

M1

$$v = \frac{12u}{25}$$

m1

Dependent on both Ms

A1F
4

(d) $v = \dfrac{2u - \frac{4}{5}u}{3} = \dfrac{2u}{5}$

M1

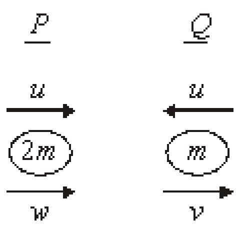
$v \succ v \Rightarrow A$ and B will not collide again

E1F
2

[14]

Q10.

(a) (i)



Conservation of momentum
$2mu - mu = mv + 2mw$

M1A1

M1 one momentum term correct

$u = v + 2w$ (1)

Restitution

$v - w = 2ue$ (2)

M1 e × speed of approach seen

M1A1

(1) – (2) gives $3w = u - 2ue$

M1

$w = \frac{u}{3}(1-2e)$

A1

(1) + 2(2) gives $u(1 + 4e) = 3v$

$v = \frac{u}{3}(1 + 4e)$

B1ft

7

(ii) v always positive, so same direction when $\frac{u}{3}(1 - 2e) > 0$

For > 0 or solving = 0

M1

$\therefore 1 - 2e > 0$

Must be convincing about <

A1

$e < \frac{1}{2}$

2

(b) (i) $I = mv - mu$

Use of mv – mu

M1

$= m\frac{u}{3}(1 + 4e) + mu$

Paired speeds correct

A1

$= 4\frac{mu}{3}(1 + e)$

Printed answer

A1

3

(ii) $0 \le e < \frac{1}{2}$

Use of e values in I

M1

$\therefore \frac{4mu}{3}(1 + 0) \le I < \frac{4mu}{3}(1 + \frac{1}{2})$

$\frac{4mu}{3} \le I < 2mu$

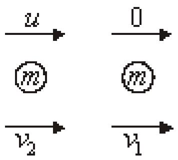
Printed answer

A1

2

[14]

Q11.



(a) Restitution: $v_1 - v_2 = eu$

Attempt at restitution

M1

Momentum: $mv_1 + mv_2 = mu$

Attempt at momentum

M1

$v_1 - v_2 = eu$ (1)

$v_1 + v_2 = eu$ (2)

Both correct

A1

(1)+(2) $2v_1 = u(1 + e)$

M1

$v_1 = \frac{u}{2}(1 - e)$

AG

A1

(2)–(1) $2v_2 = u(1 - e)$

A1

$v_2 = \frac{u}{2}(1 - e)$

B1f.t.

6

(b) Speed = $\frac{2}{3} \times \frac{u}{2}(1+e) = \frac{u}{3}(1+e)$

B1

1

(c) (i) $\frac{u}{3}(1+e) = \frac{u}{2}(1-e)$

Equating

M1

2 + 2e = 3 – 3e
5e = 1
$e = \frac{1}{5}$

Solving or showing

A1

2

(ii) Speed = $\frac{2u}{5}$

B1

1

(d) B reached wall after $\frac{5d}{3u}$

Attempt to find twice

M1A1f.t.

In this time A travels $\frac{5d}{3u} \times \frac{2u}{5} = \frac{2d}{3}$

Attempt to find distance

M1A1f.t.

A and B now have same speed so meet at
half remaining distance = $\frac{1}{2}\left(\frac{d}{3}\right)$

Attempt to find remaining distances

M1

$= \frac{d}{6}$

Special case $\frac{5d}{6} \Rightarrow$ *5 marks*

A1

6

Alternative:
Ratio of speeds after collision = 1.5 : 1

(M1A1f.t.)

Ratio of distance after collision = 1 : $\frac{2}{3}$

(M1A1f.t.)

Then $d - \frac{2}{3}d$ left = $\frac{d}{3}$

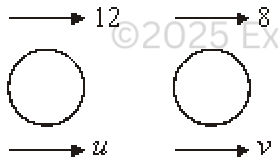
(M1)

Same speed to meet half way = $\frac{1}{2}\left(\frac{d}{3}\right) = \frac{d}{6}$

(A1)

[16]

Q12.



Cons of momentum,
12 (800) + 8 (1000) = 800 u + 1000 v

Attempt at cons of mom – 2terms correct
Fully correct – accept if 10 used

M1A1

$88 = 4u + 5v$ —(1) Restitution law,

$v - u = \frac{1}{8}(12 - 8)$

Attempt at restitution equations
Fully correct – accept if 10 used

M1 A1

$v - u = \frac{1}{2}$ —(2)

$4 \times (2) + (1)$ gives $9v = 90$

Solving a pair of simultaneous eqns

M1

$v = 10 \text{ ms}^{-1}$

A1

$\therefore u = 9.5 \text{ ms}^{-1}$

AG; v = 10 correctly obtained from pair of eqns

B1

[7]

Q13.

(a) $A \xrightarrow{43.2}$ $\qquad B \xrightarrow{0}$

$1200 \xrightarrow{v}$ $\qquad 1500 \xrightarrow{u}$

Conservation of momentum

$43.2(1200) = 1200v + 1500u$

Use of km h^{-1} or ms^{-1} throughout

M1A1

$43.2 = v + 1.25u$

(1)

Restitution Law

$u - v = \frac{1}{4}(43.2 - 0)$

M0 if 'e' on the wrong side

M1A1

$10.8 = u - v$

(2)

(1) + (2) $\qquad 54 = 2.25u$

Solving a part of equations

M1

$u = 24 \text{ km h}^{-1}$

or $6\frac{2}{3}$ *ms*$^{-1}$

A1

$v = 13.2 \text{ km h}^{-1}$

Follow through error through error in their

equations or $3\frac{2}{3}$ *ms*$^{-1}$

B1f.t.

7

(b) $24 \text{ km h}^{-1} = \dfrac{24 \times 1000}{60 \times 60}$

$= 6\frac{2}{3} \text{ ms}^{-1}$

Conversion of units correct in this part

B1

$\text{Impulse} = 1500\left(6\frac{2}{3}\right) - 0$

Impulse = change in momentum for the particle

M1

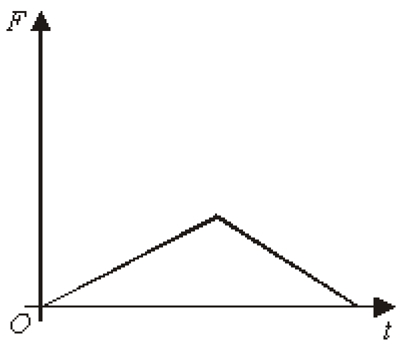
= 10 000 N s

Printed answer

A1

3

(c) (i)



B1 increase then decrease
B1 all correct

B(2,1,0)

2

(ii) Area $\Rightarrow \frac{1}{2}(0.01)F = 10000$

Use of area – uses printed answer

M1A1

$F = 2 \times 10^6$ N

Follow through error in area.

A1f.t.

3

(iii) Likely to be non-linear, or force wouldn't change abruptly

Suitable comment – dependent on graph

E1

1

[16]

Q14.

(a) Initial $\rightarrow 3u$ $\leftarrow u$

$6m$ $2m$

Final $\rightarrow V$ $\rightarrow V$

C of momentum

$6m.3u - 2m.u = 6m.V + 2m.V$

M1

$16u = 6V + 2V$

A1

Restitution $\frac{1}{4}(3u + u) = V - V$

M1A1

$u = V - V$

$14u = 8V$

M1

$V = \frac{7}{4}u$

A1

6

(b)

$V = u + \frac{7}{4}u$

M1

$= \frac{11}{4}u$

A1

2

[8]

Q15.

(a) Velocity of Q is $2u$

Impulse = $3m.2u$

M1

= $6mu$

A1

2

(b) (i) Initial $\rightarrow 2u$

Q $3m$ R $5m$

Final $\rightarrow V$

C of momentum $3m.2u = 5mV$

M1A1

$V = \frac{6}{5}u$

A1

3

(ii) Restitution $e.2u = V$

M1A1

$\therefore e = \frac{3}{5}$

A1

2

(iii) Before P collides with Q again,

P travels distance a at speed $2u$

M1

$\therefore$ Time = $\frac{a}{2u}$

A1

2

(c) After impact velocity of B is V

Conservation of momentum $mu = MV$

B1

Restitution $eu = V$

$eMu = MV = mu$

$e \leq 1$

M1

$m \leq M$

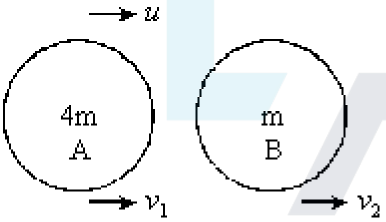
A1

3

[12]

Q16.

(a) Initial



C of Momentum: $4mu = 4mv_1 + mv_2$

M1A1

$v_2 = 2v_1$

$\Rightarrow 4u = 4v_1 + 2v_1$

M1

$\therefore\ v_1 = \frac{2}{3}u$

A1

4

(b) Restitution: $e\,u = v_2 - v_1$

M1A1

$v_2 = \frac{4}{3}u$

B1

$\therefore\ e\,u = \frac{4}{3}u - \frac{2}{3}u$

M1

$e = \frac{2}{3}$

A1

5

[9]

Q17.

(a) Impulse = change in momentum

M1

$J = 2m \times 3\mu$

$J = 6mu$

A1

2

(b) Conservation of momentum

M1

$6mu = 2mv_a + mv_b$

A1

Restitution

M1

$3u \times e = 3u \times \frac{3}{5} = v_a - v_a$

A1

$6u = 2v_a + v_b$

Subtracting $\frac{21}{5}u = 3v_a$

$\frac{7}{5}u = v_a$

A1

$6u + \frac{18}{5}u = 3v_b$

$v_b = \frac{16}{5}u$

A1

6

(c) Velocity after impact is e × velocity before impact

M1

$= \frac{1}{2} \times \frac{16}{5}u$

$= \frac{8}{5}u$

ft [from (b)]

A1ft

2

(d) Time taken for B to strike wall after collision is $\frac{4a}{\frac{16u}{5}}$

In this time A has travelled $\frac{7}{5}u \times \frac{5a}{4u}$

$= \frac{7}{4}a$

B1

Two spheres are $\frac{9}{4a}$ apart

ft [from above]

B1ft

Speeds of spheres are (A) $\frac{7}{5}u$ and (B) $\frac{8}{5}u$

A1

Distance from the wall is $\frac{8}{15} \times \frac{8}{4}a$

M1

$= \frac{6}{5}a$

A1

7

[17]

Examiner reports

Q1.

Parts of this question proved to be the most challenging on the paper, with mean marks of 59%, 47%, 13% and 32% for (a), (b), (c) and (d) respectively.

The concepts of conservation of momentum and the law of restitution were clearly familiar concepts but students made errors through lack of useful, clearly labelled diagrams. In the best solutions, students drew a diagram and labelled velocities clearly. Where labelling was not clear, it became apparent that students had confused themselves, and their momentum and restitution equations were not consistent with directions. Many students were unable to justify the printed answer for speed if they had obtained an answer of $\frac{u(2e-3)}{5}$ first.

Almost all students were able to score at least 1 mark in (b), with the most common error being in substituting the expression for speed rather than velocity.

Part (c) required an appreciation of the range of values for e in order to justify the directions of motion of the two smooth spheres. Many students failed to offer any justification and scored zero marks.

For part (d) many students were able to quote the relevant formula for impulse and to substitute corresponding expressions for velocities, but sign errors limited the number of marks available. The final result could then be obtained by considering limiting values for e. A significant number of students did not attempt this part at all.

Q2.

(a) The principle of conservation of linear momentum and the law of restitution were well applied by a great majority of students. However some students lost one or both of the last two accuracy marks by not simplifying their expressions for v_A and v_B.

(b) Here it was required to state that as $e \le 1$, then $v_B \le \frac{u}{2}(1 + 5)$, or equivalent, and hence $v_B \le 3u$. Many students gave satisfactory answers, but some lost marks due to either using a strict inequality or by not giving a fully convincing answer.

(c) Many students lost marks here for using the speed of B together with the mass of A to state the difference of the two momentums.

Q3.

Incorrect answers to part (a) of this question were very rare. However, part b(i) proved to be too challenging for some candidates. Many candidates could not answer part b(ii) correctly. The most common approach for answering part (c) was using the constant acceleration formulae; but some candidates were able to use energy methods successfully. Almost all candidates were able to state a valid modelling assumption.

Q4.

This question was attempted well, with many candidates scoring full marks. The candidates were able to use the principle of conservation of linear momentum and Newton's experimental law correctly to answer part (a). However, in part (b), a small number of candidates seemingly forgot to find the speed of B immediately after the collision. These candidates lost the last method and accuracy marks.

Part (c) proved to be too challenging for a significant number of candidates. These candidates attempted to answer this part by substituting values such as 9 or 10 for x. The more able candidates stated that $vB < 0$ for a further collision, and proceeded by solving an inequality to show the required result.

Q5.

This question proved challenging for a large number of candidates.

Many candidates were unable to provide a valid reason as to why the components of the velocities of A and B parallel to the unit vector **j** were unchanged by the collision. The response here was often "because **j** is perpendicular to **i**" or "because the spheres collide horizontally".

In part (b), a lot of candidates did not understand that the coefficient of restitution is the ratio of two speeds and not velocities. There were candidates who stated in part (a) that the components of the velocities parallel to the unit vector **j** are not changed, but then effectively ignored their own assertion when attempting to answer part (b).

Q6.

Many candidates were able to use the principle of the conservation of linear momentum and Newton's experimental law correctly to answer part (a) of this question. Many candidates were able to find the velocity of the sphere A. However, not all of these candidates were able to use the given fact that the direction of motion of the sphere A is reversed to form an inequality to answer part (b).

Even fewer candidates answered part (c) correctly. The challenge in answering the last part of the question was getting the signs of the velocities of the spheres correct as well as the direction of the inequality correct.

Q7.

The main pitfall for candidates in part (a) was a sign error in applying the principle of conservation of linear momentum. Candidates who made this error had not taken into account the opposite direction of motion of the two spheres before collision. Part (b) of this question was well-answered by the candidates.

Q8.

Part (a) This part was answered well by the great majority of candidates. However, a small number of candidates committed sign errors in using Newton's experimental law.

Part (b) Most candidates were able to apply the principle of conservation of the linear momentum and Newton's experimental law correctly but a significant number could not arrive at the correct quadratic equation in e. Most candidates solved the quadratic equation by using the formula and others by completing the square.

Q9.

Part (a) of the question was answered correctly by many candidates. Some candidates made sign errors when they applied Newton's experimental law.

For part (b), most candidates applied the Impulse/Momentum principle efficiently to the sphere B rather than to the sphere A to show the result.

In part (c), a significant number of candidates went through long algebraic manipulations to find the result, and in doing so, they made mistakes.

Candidates could not gain any marks for part (d) without first substituting $e = \frac{4}{5}$ to find the velocity of A in terms of u.

Q10.

Candidates are well versed in the use of conservation of momentum and Newton's law of restitution. Several candidates failed to take into account the sign of particle Q, thus obtaining $3mu$ for the initial momentum. Once the equations were set up they were generally solved correctly, although a few candidates forgot to find the speed of Q after the collision and careless algebraic slips sometimes cost a mark at the end. Part (a)(ii) was not well attempted with only the most able candidates being able to set up and solve the required inequality. In part (b)(i) the last mark was often dropped for not explaining why a negative sign had disappeared. This generally happened if the candidate considered the change in momentum of P rather than Q. The very last part seemed to puzzle a number of candidates who could clearly not see how the result followed by using $0 \le e < 0.5$.

Q11.

Some candidates seemed to lose the thread of this question along the way. However, a better response than expected was made to part (d).

Part (a) was done very successfully, although sometimes an error in working meant that the required speeds were interchanged. In part (c) some candidates used the restitution equation again, but to no avail.

There were a variety of responses to part (d). These included: use of distance = speed × time to build up the solution in stages; use of the total time for both spheres being the same to form a single equation and attempt to solve it; and use of ratios related to distance = speed × time. The length of the solutions varied from two lines (fully correct) to more than two pages with a minor error along the way. The last part was meant to discriminate and it largely resulted in 0, 2, 5 or 6 marks: either a start was made and not seen through or the solution was almost fully correct.

Q12.

There was an excellent response to this question. The great majority of candidates set up a pair of simultaneous equations and solved them. It was pleasing to see candidates showing sketches of the collision with velocities clearly labelled. Some candidates chose to use the answer given to find the other speed. This approach received either three or five marks depending on whether both equations had been set up or not. A few thought that Newton's law of restitution was $v + w = e(12 + 8)$,
or $(12 – 8) = e\,(v – w)$.

Q13.

The required equations were usually set up well. A mark was sometimes lost in part (a) because of a numerical slip. Candidates who used the answer given to find the speed of A

were penalised for not solving a set of simultaneous equations. Candidates who were not familiar with the correct units could not fully complete part (b). Candidates who found the impulse of B on A had to make clear reference to "equal and opposite impulses" to gain the final mark for part (b).

In part (c)(i) a few candidates thought that the graph had to be curved and not composed of two straight line segments. Part (c)(ii) proved to be a test of remembering a formula for the area of a triangle more than anything else. The quality of the criticism in part (c)(iii) showed a distinct improvement over previous papers, with candidates often indicating a sound understanding of the situation.

Q14.

This question was answered well. A few candidates made errors with signs, as initial and final diagrams were rarely shown.

Q15.

Part (a) was answered badly; many failed simply to multiply $2u$ by $3m$, the mass of Q. Part (b) was usually answered correctly. Candidates found the proof of $M \geq m$ challenging; many assumed it to be clear from $mu = Mv$. Justification was required, using $e \leq 1$, or using the fact that kinetic energy does not increase in a collision.

Q16.

A large number of candidates correctly used conservation of momentum. Those who did not use a clear 'before and after diagram' showing the directions of velocities, often used creative sign changes to obtain a positive value for e.

Q17.

Surprisingly part (a) was often incorrect; $3mu$ was a common answer. Again, in part (b) for those candidates who did not use a diagram, signs were often incorrect, particularly in the equation using coefficient of restitution. Those who did use the correct signs had little difficulty in parts (b) and (c). The majority of those who appreciated what was happening in part (d) were also successful. Many found the position of A when B hit the wall. Using the ratio of the final speeds was found to be a simple method of finding the distance of the collision from the wall.